From Bits to Pixels

The Mathematical and Physical Foundations of elementary 3D Graphics

Jack Wetherell

Contents

1 Foundations: Numbers, Memory, and Representation **3**
1.1 Bits, Bytes, and Binary Storage . . . 3
1.2 Floating Point Representation (IEEE 754) . . . 3
1.3 From Raw Memory to Numerical Values . . . 4

2 Geometry Representation in Memory **5**
2.1 Vertices (Position Vectors) . . . 5
2.2 Triangles (Faces) . . . 5
2.3 Indexed vs Non-Indexed Geometry . . . 6
2.4 Meshes (Objects) . . . 6
2.5 Scenes as Collections of Objects . . . 7

3 Basis Vectors and Coordinate Consistency **7**
3.1 Implicit Basis in Matrix Representations . . . 7
3.2 Consistency of Basis in Matrix Multiplication . . . 8
3.3 Local vs World Basis Interpretation . . . 8
3.4 Transform Chains as Basis Transitions . . . 9
3.5 Conceptual Summary . . . 9

4 Coordinate Systems and Spaces **9**
4.1 Local (Object) Space . . . 10
4.2 World Space . . . 10
4.3 Camera (View) Space . . . 10
4.4 Clip Space . . . 11
4.5 Normalized Device Coordinates (NDC) . . . 11
4.6 Screen Space . . . 12
4.7 Summary of Transform Pipeline . . . 12
4.8 Conceptual Interpretation . . . 12

5 Rasterization **12**
5.1 Triangle Setup . . . 13
5.2 Barycentric Coordinates . . . 13
5.3 Interpolation Across Surfaces . . . 13
5.4 Depth Testing (Z-Buffer) . . . 14
5.5 Summary . . . 14

6 Memory Layout of Graphics Data **15**
6.1 Vertex Buffers . . . 15
6.2 Index Buffers . . . 15
6.3 Transformation Storage . . . 16
6.4 Estimating Memory Usage . . . 16
6.5 Example: Pyramid . . . 17
6.6 Summary . . . 17

7 Specialized Hardware in GPUs **17**
7.1 Graphics Pipelines and Fixed Function Units . . . 17
7.2 Ray Tracing Cores (RT Cores) . . . 18
7.3 Tensor Cores and Machine Learning . . . 19
7.4 Evolution of GPU Specialization . . . 19
7.5 Conceptual Interpretation . . . 20

1 Foundations: Numbers, Memory, and Representation

1.1 Bits, Bytes, and Binary Storage

At the lowest level, all data in a computer is represented using binary digits.

- A **bit** is a single binary value:
$$b \in \{0, 1\} \tag{1}$$
- A **byte** consists of 8 bits:
$$1 \text{ byte} = 8 \text{ bits} \tag{2}$$

A sequence of bits can represent integers using positional notation in base 2:

$$N = \sum_{i=0}^{n-1} b_i \cdot 2^i \tag{3}$$

where:

- b_i is the i-th bit (0 or 1)
- $i = 0$ corresponds to the least significant bit (LSB)

Example:
$$1011_2 = 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 11_{10} \tag{4}$$

Memory is therefore a linear array of bytes:

$$\text{Memory} = [\text{byte}_0, \text{byte}_1, \ldots, \text{byte}_N] \tag{5}$$

Each byte has an address:

$$\text{Address} \in \mathbb{N} \tag{6}$$

1.2 Floating Point Representation (IEEE 754)

Real numbers are represented approximately using the IEEE 754 floating point format.

A 32-bit floating point number (single precision) is composed of:

- 1 bit for the sign (s)
- 8 bits for the exponent (e)
- 23 bits for the mantissa (fraction) (m)

The value represented is:

$$x = (-1)^s \cdot (1 + m) \cdot 2^{e-127} \tag{7}$$

where:

$$m = \sum_{i=1}^{23} m_i \cdot 2^{-i} \tag{8}$$

Interpretation:

- $(-1)^s$ determines the sign
- $(1 + m)$ is the normalized significand
- $e - 127$ is the unbiased exponent

Special Cases:

- $e = 0 \rightarrow$ denormalized numbers
- $e = 255 \rightarrow$ infinity or NaN

1.3 From Raw Memory to Numerical Values

All numerical values in a program are ultimately stored as sequences of bytes in memory.

A 32-bit float occupies 4 consecutive bytes:

$$\text{float} \rightarrow [b_0, b_1, b_2, b_3] \tag{9}$$

These bytes are interpreted according to the IEEE 754 structure:

$$\underbrace{s}_{1\text{ bit}} \quad \underbrace{e}_{8\text{ bits}} \quad \underbrace{m}_{23\text{ bits}} \tag{10}$$

Example: Representing $x = 7.98$ The decimal number 7.98 cannot be represented exactly in binary, so it is approximated.

Its IEEE 754 single-precision representation is:

$$x \approx 7.980000019073486 \tag{11}$$

Binary layout:

$$0\ 10000001\ 11111111001100110011010 \tag{12}$$

Breaking this down:

- Sign bit:
$$s = 0 \quad \text{(positive)} \tag{13}$$
- Exponent:
$$e = 10000001_2 = 129_{10} \tag{14}$$
$$e_{\text{unbiased}} = 129 - 127 = 2 \tag{15}$$
- Mantissa:
$$m = 1.11111111001100110011010_2 \tag{16}$$

Reconstructing the value:

$$x = (-1)^0 \cdot (1 + m_{\text{fraction}}) \cdot 2^2 \tag{17}$$

Key Insight: There is no intrinsic meaning in memory—only interpretation.

$$\text{Bits} \xrightarrow{\text{interpretation}} \text{Number} \tag{18}$$

The same sequence of bits may represent:

- an integer
- a floating point number
- part of a vertex
- or any structured data

Conclusion: All higher-level constructs (vectors, matrices, geometry) are built from:

$$\text{Bits} \to \text{Bytes} \to \text{Floats} \to \text{Mathematical Objects} \tag{19}$$

2 Geometry Representation in Memory

2.1 Vertices (Position Vectors)

A **vertex** represents a point in 3D space.

In its simplest form, a vertex consists of three floating-point values:

$$\mathbf{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \tag{20}$$

For transformation purposes, vertices are often expressed in homogeneous coordinates:

$$\mathbf{v}_h = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \in \mathbb{R}^4 \tag{21}$$

Memory Representation: Each coordinate is a 32-bit float:

$$\text{vertex} = [x, y, z] \tag{22}$$

$$\text{size of vertex} = 3 \times 4 \text{ bytes} = 12 \text{ bytes} \tag{23}$$

Thus, a vertex buffer is:

$$V = [\mathbf{v}_0, \mathbf{v}_1, \ldots, \mathbf{v}_n] \tag{24}$$

2.2 Triangles (Faces)

The fundamental rendering primitive is the **triangle**.

A triangle is defined by three vertices:

$$\triangle = (\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2) \tag{25}$$

Geometrically, any point inside the triangle can be expressed using barycentric coordinates:

$$\mathbf{p} = \alpha\mathbf{v}_0 + \beta\mathbf{v}_1 + \gamma\mathbf{v}_2 \tag{26}$$

subject to:

$$\alpha + \beta + \gamma = 1, \quad \alpha, \beta, \gamma \geq 0 \tag{27}$$

Memory Representation (Non-Indexed): Each triangle stores three full vertices:

$$\text{triangle size} = 3 \times 12 \text{ bytes} = 36 \text{ bytes} \tag{28}$$

2.3 Indexed vs Non-Indexed Geometry

Non-Indexed Representation: Each triangle explicitly stores its vertices:

$$T = [\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2] \tag{29}$$

This leads to duplication of shared vertices.

Indexed Representation: Vertices are stored once in a vertex buffer:

$$V = [\mathbf{v}_0, \mathbf{v}_1, \ldots, \mathbf{v}_n] \tag{30}$$

Triangles are defined using indices:

$$F = [(i_0, i_1, i_2), (i_3, i_4, i_5), (i_6, i_7, i_8)] \tag{31}$$

Where winding order is used to specify face direction.
Each index refers to a vertex:

$$\triangle_k = (\mathbf{v}_{i_a}, \mathbf{v}_{i_b}, \mathbf{v}_{i_c}) \tag{32}$$

Memory Efficiency: Let:

$$n_v = \text{number of vertices}, \quad n_t = \text{number of triangles} \tag{33}$$

Then:

$$\text{non-indexed size} = 36 \cdot n_t \tag{34}$$

$$\text{indexed size} = 12 \cdot n_v + 12 \cdot n_t \tag{35}$$

(assuming 32-bit indices)

2.4 Meshes (Objects)

A **mesh** (object) consists of:

- A vertex buffer V
- An index buffer F

$$\mathcal{M} = (V, F) \tag{36}$$

The mesh is defined in **local (object) space**, meaning:

$$\mathbf{v}_{local} \in \mathbb{R}^3 \tag{37}$$

All vertices are expressed relative to the object's origin:

$$\mathbf{o}_{local} = (0, 0, 0) \tag{38}$$

2.5 Scenes as Collections of Objects

A **scene** consists of multiple objects:

$$\mathcal{S} = \{\mathcal{M}_1, \mathcal{M}_2, \dots, \mathcal{M}_k\} \tag{39}$$

Each object has an associated transformation:

$$M_i \in \mathbb{R}^{4\times 4} \tag{40}$$

which maps local coordinates to world coordinates:

$$\mathbf{v}_{world} = M_i \mathbf{v}_{local} \tag{41}$$

Global Interpretation: All objects share a common coordinate system:

$$\mathbf{v}_{world} \in \mathbb{R}^3 \tag{42}$$

Thus, the scene becomes a unified collection of geometry in world space.

Final Structure in Memory:

$$\text{Scene} = \{(V_1, F_1, M_1), (V_2, F_2, M_2), \dots\} \tag{43}$$

Each component ultimately reduces to:

$$\text{Bytes} \to \text{Floats} \to \text{Vertices} \to \text{Triangles} \to \text{Meshes} \to \text{Scene} \tag{44}$$

3 Basis Vectors and Coordinate Consistency

All vector and matrix operations in graphics implicitly depend on a chosen basis. Understanding how basis vectors are represented and transformed is essential for interpreting coordinate systems and transformation pipelines.

3.1 Implicit Basis in Matrix Representations

Any vector is expressed relative to a basis.

Let a basis be:

$$\mathcal{B} = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\} \tag{45}$$

Then a vector $\mathbf{v}$ is written as:

$$\mathbf{v} = x\mathbf{e}_1 + y\mathbf{e}_2 + z\mathbf{e}_3 \tag{46}$$

In coordinate form:

$$[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \tag{47}$$

Key Insight: The components (x, y, z) have meaning only with respect to a basis.

$$\text{Coordinates} \neq \text{Vector} \quad \text{(without basis)} \tag{48}$$

Matrices Encode Bases: A matrix can be interpreted as a set of basis vectors:

$$M = \begin{bmatrix} | & | & | \\ \mathbf{c}_1 & \mathbf{c}_2 & \mathbf{c}_3 \\ | & | & | \end{bmatrix} \tag{49}$$

where each column $\mathbf{c}_i$ represents a transformed basis vector.
Thus:

$$M\mathbf{e}_i = \mathbf{c}_i \tag{50}$$

3.2 Consistency of Basis in Matrix Multiplication

Matrix multiplication is only valid when all operands are expressed in a consistent basis.
Given:

$$\mathbf{v}' = M\mathbf{v} \tag{51}$$

this implies:

$$[\mathbf{v}]_{\mathcal{B}} \to [\mathbf{v}']_{\mathcal{B}} \tag{52}$$

Both $\mathbf{v}$ and M must be defined relative to the same basis $\mathcal{B}$.

Composition of Transformations: For two transformations:

$$\mathbf{v}'' = M_2 M_1 \mathbf{v} \tag{53}$$

the intermediate result:

$$\mathbf{v}' = M_1 \mathbf{v} \tag{54}$$

must be expressed in the same coordinate system expected by M_2.

Key Constraint:

$$M_2 \cdot M_1 \quad \text{is valid only if their bases are aligned} \tag{55}$$

3.3 Local vs World Basis Interpretation

In graphics, different spaces correspond to different bases.

Local Basis:

$$\mathcal{B}_{local} = \{\mathbf{e}_x^{local}, \mathbf{e}_y^{local}, \mathbf{e}_z^{local}\} \tag{56}$$

Vertices are expressed as:

$$[\mathbf{v}]_{local} \tag{57}$$

World Basis:

$$\mathcal{B}_{world} = \{\mathbf{e}_x^{world}, \mathbf{e}_y^{world}, \mathbf{e}_z^{world}\} \tag{58}$$

The model matrix maps between these:

$$\mathbf{v}_{world} = M_{model}\mathbf{v}_{local} \tag{59}$$

Interpretation of M_{model}:

$$M_{model} = \begin{bmatrix} | & | & | \\ \mathbf{e}_x^{local \to world} & \mathbf{e}_y^{local \to world} & \mathbf{e}_z^{local \to world} \\ | & | & | \end{bmatrix} \tag{60}$$

Each column describes how a local basis vector is expressed in world coordinates.

3.4 Transform Chains as Basis Transitions

The full graphics pipeline can be interpreted as a sequence of basis changes:

$$\mathbf{v}_{clip} = M_{proj} \cdot M_{view} \cdot M_{model} \cdot \mathbf{v}_{local} \tag{61}$$

Each matrix performs a change of basis:

- M_{model}: local $\to$ world
- M_{view}: world $\to$ camera
- M_{proj}: camera $\to$ clip

Thus:

$$[\mathbf{v}]_{local} \to [\mathbf{v}]_{world} \to [\mathbf{v}]_{camera} \to [\mathbf{v}]_{clip} \tag{62}$$

Key Insight: Matrix multiplication composes basis transformations:

$$M_{total} = M_{proj} \cdot M_{view} \cdot M_{model} \tag{63}$$

so that:

$$\mathbf{v}_{clip} = M_{total}\mathbf{v}_{local} \tag{64}$$

3.5 Conceptual Summary

- Every vector is defined relative to a basis
- Every matrix encodes a transformation of basis vectors
- Matrix multiplication composes consistent basis transformations
- The graphics pipeline is a sequence of basis changes

$$\text{Coordinates} + \text{Basis} = \text{Geometric Meaning} \tag{65}$$

$$\text{Transformations} = \text{Changes of Basis} \tag{66}$$

4 Coordinate Systems and Spaces

In computer graphics, geometry is transformed through a sequence of coordinate systems. Each space provides a different frame of reference for interpreting vertex positions.

$$\mathbf{v}_{local} \to \mathbf{v}_{world} \to \mathbf{v}_{camera} \to \mathbf{v}_{clip} \to \mathbf{v}_{ndc} \to \mathbf{v}_{screen} \tag{67}$$

4.1 Local (Object) Space

Local space (or object space) is the coordinate system in which a mesh is originally defined.

$$\mathbf{v}_{local} = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \tag{68}$$

Key properties:

- Origin is the object's center:
$$\mathbf{o}_{local} = (0, 0, 0) \tag{69}$$
- Basis vectors define the object's orientation:
$$\{\mathbf{e}_x^{local}, \mathbf{e}_y^{local}, \mathbf{e}_z^{local}\} \tag{70}$$

Vertices are stored in this space in memory.

4.2 World Space

World space is a global coordinate system shared by all objects in the scene.

Each object is placed into world space using a model transformation:

$$\mathbf{v}_{world} = M_{model}\mathbf{v}_{local} \tag{71}$$

where:

$$M_{model} = T \cdot R \cdot S \tag{72}$$

Explicitly:

$$M_{model} = \begin{bmatrix} r_{11}S_x & r_{12}S_y & r_{13}S_z & t_x \\ r_{21}S_x & r_{22}S_y & r_{23}S_z & t_y \\ r_{31}S_x & r_{32}S_y & r_{33}S_z & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{73}$$

Key properties:

- Common reference frame for all objects
- Fixed global basis:
$$\{\mathbf{e}_x^{world}, \mathbf{e}_y^{world}, \mathbf{e}_z^{world}\} \tag{74}$$

4.3 Camera (View) Space

Camera space (or view space) represents coordinates relative to the camera.

Instead of moving the camera, we transform the world:

$$\mathbf{v}_{camera} = M_{view}\mathbf{v}_{world} \tag{75}$$

The view matrix is the inverse of the camera transform:

$$M_{view} = (T_{cam}R_{cam})^{-1} \tag{76}$$

If the camera has position $\mathbf{c}$ and rotation R_{cam}:

$$M_{view} = \begin{bmatrix} R_{cam}^T & -R_{cam}^T\mathbf{c} \\ 0 & 1 \end{bmatrix} \tag{77}$$

Key properties:

- Camera is located at origin:

$$\mathbf{o}_{camera} = (0, 0, 0) \tag{78}$$

- Viewing direction typically along negative z-axis

4.4 Clip Space

Clip space is obtained by applying the projection matrix:

$$\mathbf{v}_{clip} = M_{proj}\mathbf{v}_{camera} \tag{79}$$

A standard perspective projection matrix is:

$$M_{proj} = \begin{bmatrix} \frac{1}{a\tan(\frac{\text{fov}}{2})} & 0 & 0 & 0 \\ 0 & \frac{1}{\tan(\frac{\text{fov}}{2})} & 0 & 0 \\ 0 & 0 & \frac{f+n}{n-f} & \frac{2fn}{n-f} \\ 0 & 0 & -1 & 0 \end{bmatrix} \tag{80}$$

where:

$$a = \text{aspect ratio}, \quad n = \text{near plane}, \quad f = \text{far plane} \quad \text{fov} = \text{field of view} \tag{81}$$

Vertices in clip space are 4D:

$$\mathbf{v}_{clip} = \begin{bmatrix} x_c \\ y_c \\ z_c \\ w_c \end{bmatrix} \tag{82}$$

4.5 Normalized Device Coordinates (NDC)

To obtain normalized device coordinates, perform the perspective divide:

$$\mathbf{v}_{ndc} = \begin{bmatrix} x_c/w_c \\ y_c/w_c \\ z_c/w_c \end{bmatrix} \tag{83}$$

This maps coordinates into a canonical cube:

$$x, y, z \in [-1, 1] \tag{84}$$

Key properties:

- Removes perspective distortion via division by w
- Enables uniform clipping and rasterization

4.6 Screen Space

Screen space maps normalized coordinates to pixel coordinates.

For a viewport of width W and height H:

$$x_{screen} = \frac{W}{2}(x_{ndc} + 1) \tag{85}$$

$$y_{screen} = \frac{H}{2}(1 - y_{ndc}) \tag{86}$$

$$z_{screen} = \frac{1}{2}(z_{ndc} + 1) \tag{87}$$

Final screen coordinate:

$$\mathbf{v}_{screen} = \begin{bmatrix} x_{screen} \\ y_{screen} \\ z_{screen} \end{bmatrix} \tag{88}$$

4.7 Summary of Transform Pipeline

The full transformation pipeline is:

$$\mathbf{v}_{clip} = M_{proj} \cdot M_{view} \cdot M_{model} \cdot \mathbf{v}_{local} \tag{89}$$

Followed by:

$$\mathbf{v}_{ndc} = \frac{1}{w}\mathbf{v}_{clip} \tag{90}$$

$$\mathbf{v}_{screen} = \text{Viewport}(\mathbf{v}_{ndc}) \tag{91}$$

4.8 Conceptual Interpretation

Each transformation corresponds to a change of coordinate system:

$$\text{Local} \to \text{World} \to \text{Camera} \to \text{Clip} \to \text{Screen} \tag{92}$$

At every stage:

- Coordinates are expressed relative to a specific basis
- Matrix multiplication re-expresses vectors in a new basis

Thus, the entire graphics pipeline can be viewed as a sequence of basis transformations applied to vertex data.

5 Rasterization

Rasterization is the process of converting geometric primitives (triangles) into discrete pixels on the screen.

Given vertices in screen space:

$$\mathbf{v}_0 = (x_0, y_0, z_0), \quad \mathbf{v}_1 = (x_1, y_1, z_1), \quad \mathbf{v}_2 = (x_2, y_2, z_2) \tag{93}$$

the goal is to determine which pixels lie inside the triangle and compute their attributes.

5.1 Triangle Setup

A triangle is defined by three vertices in screen space:

$$\triangle = (\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2) \tag{94}$$

We define edge functions to determine whether a point lies inside the triangle.
For a point $\mathbf{p} = (x, y)$, define:

$$E_{ij}(\mathbf{p}) = (x - x_i)(y_j - y_i) - (y - y_i)(x_j - x_i) \tag{95}$$

A point is inside the triangle if:

$$E_{01}(\mathbf{p}) \geq 0, \quad E_{12}(\mathbf{p}) \geq 0, \quad E_{20}(\mathbf{p}) \geq 0 \tag{96}$$

(for a consistently wound triangle)

Interpretation: Each edge function represents a signed area test.

5.2 Barycentric Coordinates

Any point inside the triangle can be expressed as a weighted combination of the vertices:

$$\mathbf{p} = \alpha \mathbf{v}_0 + \beta \mathbf{v}_1 + \gamma \mathbf{v}_2 \tag{97}$$

with:

$$\alpha + \beta + \gamma = 1 \tag{98}$$

The barycentric coordinates can be computed using areas:

$$\alpha = \frac{A(\mathbf{p}, \mathbf{v}_1, \mathbf{v}_2)}{A(\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2)} \tag{99}$$

$$\beta = \frac{A(\mathbf{v}_0, \mathbf{p}, \mathbf{v}_2)}{A(\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2)} \tag{100}$$

$$\gamma = \frac{A(\mathbf{v}_0, \mathbf{v}_1, \mathbf{p})}{A(\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2)} \tag{101}$$

where:

$$A(\mathbf{a}, \mathbf{b}, \mathbf{c}) = \frac{1}{2} \begin{vmatrix} a_x & a_y & 1 \\ b_x & b_y & 1 \\ c_x & c_y & 1 \end{vmatrix} \tag{102}$$

5.3 Interpolation Across Surfaces

Once barycentric coordinates are known, any vertex attribute can be interpolated.
Let q be a per-vertex attribute (e.g. depth, color, texture coordinate):

$$q_0, \quad q_1, \quad q_2 \tag{103}$$

Then:

$$q(\mathbf{p}) = \alpha q_0 + \beta q_1 + \gamma q_2 \tag{104}$$

Important: Perspective-Correct Interpolation Due to perspective projection, interpolation must be corrected using w:

$$q' = \frac{q}{w} \tag{105}$$

Interpolate:

$$q'(\mathbf{p}) = \alpha q'_0 + \beta q'_1 + \gamma q'_2 \tag{106}$$

Then recover:

$$q(\mathbf{p}) = \frac{q'(\mathbf{p})}{\alpha \frac{1}{w_0} + \beta \frac{1}{w_1} + \gamma \frac{1}{w_2}} \tag{107}$$

5.4 Depth Testing (Z-Buffer)

Each pixel stores a depth value:

$$z_{\text{buffer}}(x, y) \tag{108}$$

For a candidate fragment with depth z_{new}:

$$z_{\text{new}} < z_{\text{buffer}}(x, y) \Rightarrow \text{fragment is visible} \tag{109}$$

If true:

$$z_{\text{buffer}}(x, y) \leftarrow z_{\text{new}} \tag{110}$$

$$\text{color}(x, y) \leftarrow \text{fragment color} \tag{111}$$

Otherwise, the fragment is discarded.

Depth Range: After projection:

$$z \in [0, 1] \tag{112}$$

Non-linearity: Depth is non-linear with respect to camera distance:

$$z \sim \frac{1}{z_{camera}} \tag{113}$$

This provides higher precision near the camera.

5.5 Summary

Rasterization converts triangles into pixels through:

$$\text{Triangle} \rightarrow \text{Fragments} \rightarrow \text{Pixels} \tag{114}$$

Key steps:

- Determine pixel coverage (edge functions)
- Compute barycentric coordinates
- Interpolate attributes
- Perform depth testing

Thus, rasterization bridges continuous geometry and discrete image representation.

6 Memory Layout of Graphics Data

All geometric data in a graphics pipeline is ultimately stored as structured sequences of bytes in memory (CPU RAM or GPU VRAM). These structures are designed for efficient access and parallel processing.

6.1 Vertex Buffers

A **vertex buffer** stores a sequence of vertices:

$$V = [\mathbf{v}_0, \mathbf{v}_1, \ldots, \mathbf{v}_{n-1}] \tag{115}$$

Each vertex typically contains multiple attributes:

$$\mathbf{v}_i = \Big[x_i, y_i, z_i, w_i,\ n_x, n_y, n_z,\ u, v,\ \ldots\Big] \tag{116}$$

Memory Layout: Each attribute is stored as a float (4 bytes):

$$\text{size of vertex} = k \cdot 4 \text{ bytes} \tag{117}$$

where k is the number of float components.

Example (Position Only):

$$\mathbf{v}_i = (x, y, z) \tag{118}$$

$$\text{vertex size} = 3 \times 4 = 12 \text{ bytes} \tag{119}$$

Contiguous Storage:

$$\text{Vertex Buffer} = [x_0, y_0, z_0,\ x_1, y_1, z_1,\ \ldots] \tag{120}$$

6.2 Index Buffers

An **index buffer** defines how vertices form triangles.

$$F = [(i_0, i_1, i_2), (i_3, i_4, i_5), \ldots] \tag{121}$$

Each index refers to a vertex in the vertex buffer:

$$\triangle_k = (\mathbf{v}_{i_a}, \mathbf{v}_{i_b}, \mathbf{v}_{i_c}) \tag{122}$$

Memory Representation: Indices are typically stored as integers:

$$\text{index size} = \begin{cases} 2 \text{ bytes (16-bit)} \\ 4 \text{ bytes (32-bit)} \end{cases} \tag{123}$$

Example:

$$F = [0, 1, 2,\ 2, 3, 0] \tag{124}$$

This represents two triangles sharing vertices.

6.3 Transformation Storage

Each object in a scene has an associated transformation matrix:

$$M \in \mathbb{R}^{4\times 4} \tag{125}$$

Matrix Layout:

$$M = \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{32} & m_{33} & m_{34} \\ m_{41} & m_{42} & m_{43} & m_{44} \end{bmatrix} \tag{126}$$

Memory Size:

$$\text{matrix size} = 16 \times 4 = 64 \text{ bytes} \tag{127}$$

Storage Format: Matrices are stored as a flat array:

$$[m_{11}, m_{12}, \ldots, m_{44}] \tag{128}$$

Note: Different APIs may use:

- Row-major order
- Column-major order

6.4 Estimating Memory Usage

Let:

$$n_v = \text{number of vertices} \tag{129}$$

$$n_t = \text{number of triangles} \tag{130}$$

$$k = \text{floats per vertex} \tag{131}$$

Vertex Buffer Size:

$$\text{size}_V = n_v \cdot k \cdot 4 \tag{132}$$

Index Buffer Size: Each triangle uses 3 indices:

$$\text{size}_F = n_t \cdot 3 \cdot s_i \tag{133}$$

where:

$$s_i \in \{2, 4\} \text{ bytes} \tag{134}$$

Transformation Size: For n_o objects:

$$\text{size}_M = n_o \cdot 64 \tag{135}$$

Total Memory:

$$\text{Total} = \text{size}_V + \text{size}_F + \text{size}_M \tag{136}$$

6.5 Example: Pyramid

Consider a pyramid with:

- $n_v = 5$ vertices
- $n_t = 6$ triangles
- $k = 3$ (position only)

Vertex Memory:

$$\text{size}_V = 5 \cdot 3 \cdot 4 = 60 \text{ bytes} \tag{137}$$

Index Memory (32-bit):

$$\text{size}_F = 6 \cdot 3 \cdot 4 = 72 \text{ bytes} \tag{138}$$

Transform:

$$\text{size}_M = 64 \text{ bytes} \tag{139}$$

Total:

$$\text{Total} = 60 + 72 + 64 = 196 \text{ bytes} \tag{140}$$

6.6 Summary

Graphics data in memory is structured as:

$$\text{Scene} = \{(V_i, F_i, M_i)\}_{i=1}^{n_o} \tag{141}$$

Each component is stored as contiguous bytes:

$$\text{Bytes} \to \text{Floats} \to \text{Vertices} \to \text{Triangles} \to \text{Objects} \tag{142}$$

Efficient layout enables:

- Coalesced memory access on GPU
- Parallel processing of vertices
- High throughput rendering

7 Specialized Hardware in GPUs

Modern GPUs are highly specialized processors designed to execute large numbers of similar operations in parallel. Over time, they have evolved from fixed-function graphics devices to programmable, massively parallel compute units with dedicated hardware for specific tasks.

7.1 Graphics Pipelines and Fixed Function Units

Early GPUs implemented a **fixed-function pipeline**, where each stage of the rendering process was hardwired.

The classical pipeline can be represented as:

$$\mathbf{v}_{clip} = M_{proj} \cdot M_{view} \cdot M_{model} \cdot \mathbf{v}_{local} \tag{143}$$

followed by rasterization:

$$\text{Triangles} \to \text{Fragments} \to \text{Pixels} \tag{144}$$

Fixed-Function Stages:

- Vertex transformation
- Primitive assembly
- Clipping
- Rasterization
- Depth testing

These stages were implemented directly in hardware, meaning:

$$\text{Transformation} = \text{Dedicated combinational logic} \tag{145}$$

Modern GPUs replace most fixed-function stages with programmable shaders:

$$\text{Vertex Shader}, \quad \text{Fragment Shader}, \quad \text{Compute Shader} \tag{146}$$

However, some stages (e.g., rasterization, depth testing) remain hardware-accelerated.

7.2 Ray Tracing Cores (RT Cores)

Ray tracing computes visibility by tracing rays through a scene.

A ray is defined as:

$$\mathbf{r}(t) = \mathbf{o} + t\mathbf{d} \tag{147}$$

where:

- $\mathbf{o}$ is the ray origin
- $\mathbf{d}$ is the direction
- $t \geq 0$

The core operation is ray-geometry intersection:

$$\mathbf{r}(t) = \text{surface} \tag{148}$$

For triangles, this involves solving:

$$\mathbf{o} + t\mathbf{d} = \alpha\mathbf{v}_0 + \beta\mathbf{v}_1 + \gamma\mathbf{v}_2 \tag{149}$$

subject to:

$$\alpha + \beta + \gamma = 1 \tag{150}$$

RT cores accelerate:

- Ray-triangle intersection
- Ray traversal through acceleration structures (e.g., BVH)

This reduces computational complexity from:

$$O(n) \to O(\log n) \tag{151}$$

for scene traversal.

7.3 Tensor Cores and Machine Learning

Tensor cores are specialized units designed for matrix operations.

The fundamental operation is matrix multiplication:

$$C = A \cdot B \tag{152}$$

with:

$$C_{ij} = \sum_k A_{ik} B_{kj} \tag{153}$$

These operations are central to:

- Neural networks
- Linear algebra
- Physics simulations

Tensor cores perform fused operations:

$$D = A \cdot B + C \tag{154}$$

This is known as a **fused multiply-add (FMA)**.

Key Property: These units execute many operations in parallel:

$$\text{Throughput} \gg \text{CPU scalar performance} \tag{155}$$

7.4 Evolution of GPU Specialization

The evolution of GPUs can be understood as increasing specialization of hardware for common computational patterns.

Stage 1: Fixed-Function Graphics

$$\text{Hardwired pipeline for rendering} \tag{156}$$

Stage 2: Programmable Shaders

$$\text{Generalized arithmetic units (ALUs)} \tag{157}$$

Stage 3: General-Purpose GPU (GPGPU)

$$\text{Parallel execution of arbitrary compute kernels} \tag{158}$$

Stage 4: Specialized Accelerators

- RT cores (ray tracing)
- Tensor cores (matrix operations)

$$\text{Specialized hardware} \Rightarrow \text{higher efficiency} \tag{159}$$

7.5 Conceptual Interpretation

All GPU operations ultimately reduce to:

$$\text{Logic Gates} \rightarrow \text{Arithmetic Units} \rightarrow \text{Matrix Operations} \tag{160}$$

Thus:

$$\text{Graphics} \approx \text{Linear Algebra} \approx \text{Parallel Computation} \tag{161}$$

The GPU is therefore best understood as:

$$\text{A massively parallel linear algebra processor with specialized accelerators} \tag{162}$$